Steps to connect to Database

1. Register driver

DriverManager.registerDriver(new com.mysql.cj.jdbc.Driver());
Class.forName("com.mysql.cj.jdbc.Driver")

2. Create connection

   Connection conn=DriverManager.getConnection(url,username,password);

String url="jdbc:mysql://localhost:3306/test?useSSL=false
&allowPublicKeyRetrieval=true"

   Connection conn=DriverManager.getConnection(url,"root","root1234");

3. Create statement

   Statements are of 3 types

   1. Statement
      a. This class object is used to fire query, if the query is changing every time
      b. It is very vulnerable to SQL Injection, less secure
      c. To create statement
         Statement st=conn.createStatement();
         st.executeQuery("select * from emp");
         st.executeQuery("select * from product")
   2. PreparedStatement
      a. This class object is used to fire query, if the query is same every time, but data is changing
      b. This statement store execution plan, and reuse it everytime.
      c. It is very secure statement, it avoid SQL injection vulnerability

      PreparedStatement pst=conn.prepareStatement("select * from user where uname=?");

   3. CallableStatement
      a. It is mainly used to call functions and procedures in Database

   .CallableStatement cst=comm.prepareCall("{call getCnt(?,?)}")

4. Execute query

   To execute query there are 3 functions

   1. executeQuery -→ It is used for executing select statement, it returns ResultSet
   2. executeUpdate--→ it is used to execute all DML statement, it returns an integer
   3. execute-→ it is used to call procedures /functions, it returns true/false

To use it with Statement object
Statement st=conn.createStatement();
ResultSet rs=st.executeQuery("select * from Product")

```
while(rs.next()){
  System.out.println("Id: "+rs.getInt(1)+" Name : "+rs.getString(2)+" Qty: "+rs.getInt(3));
}
```

```
Int pid=3;
String name="lays";
Int qty=34;

int n=st.executeUpdate("insert into product values("+pid+",'"+name+"',"+qty+")")
if(n>0){
   System.out.println("insertion done");
}else{
   System.out.println("insertion not done");
}
```

Using Prepared statement

```
PreparedStatement pst=conn.prepareStatement("select * from Product");
ResultSet rs=pst.executeQuery()
while(rs.next()){
  System.out.println("Id: "+rs.getInt(1)+" Name : "+rs.getString(2)+" Qty: "+rs.getInt(3));
}
```

```
Int pid=3;
String name="lays";
Int qty=34;
PreparedStatement pst1=conn.prepareStatement("insert into product values(?,?,?)");
pst1.setInt(1,pid)
pst1.setString(2,name);
pst1.setInt(3,qty);
int n=pst1.executeUpdate()
if(n>0){
   System.out.println("insertion done");
}else{
   System.out.println("insertion not done");
}
```

Using Callable statement

```
Int num=20;
CallableStatement cst=conn.prepareCall("{call getcnt(?,?)}");
cst.setInt(1,num);
cst.registerOutParameter(2,Types.Integer);

boolean ans=cst.execute();
System.out.println("count of dept "+num + "---->"+cst.getInt(2));
```

Connection, PreparedSatement, CallableStatement, Statement all these are Interfaces, Its implementation classes are available in Jar files, which we add as external jar in our program

And functions like getConnection, prepareStatement,createStatement are using factory pattern, because all these functions create object of different class.